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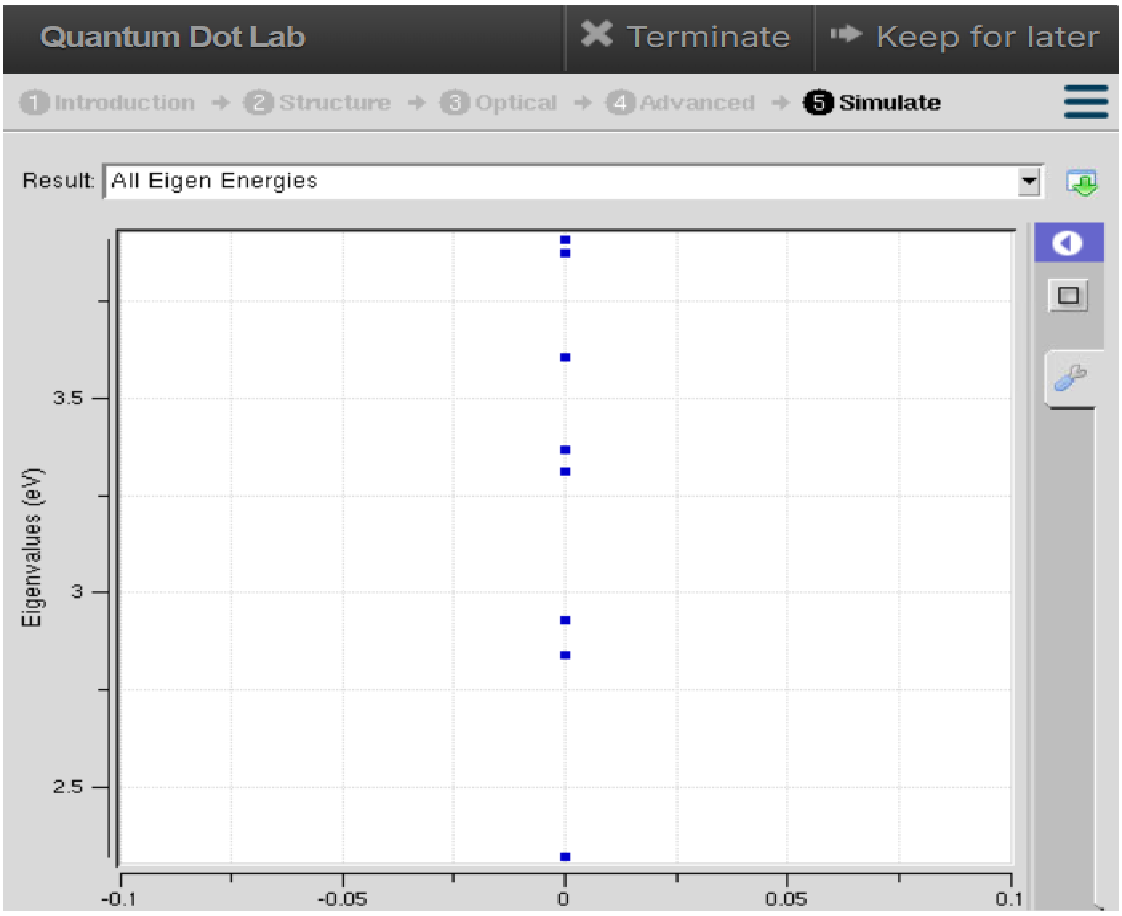
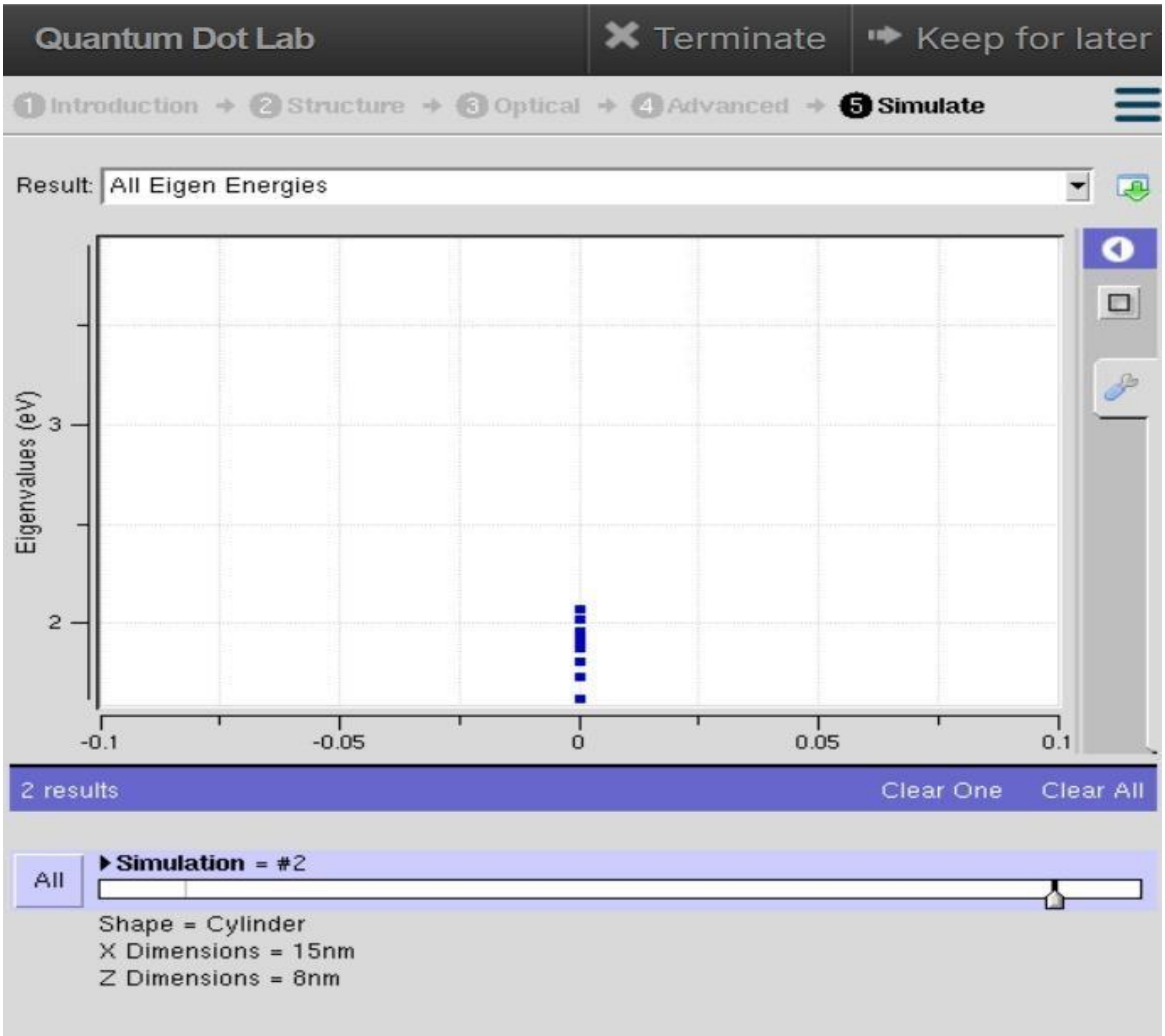
Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

Aim:
To obtain the energy states of Quantum dots.



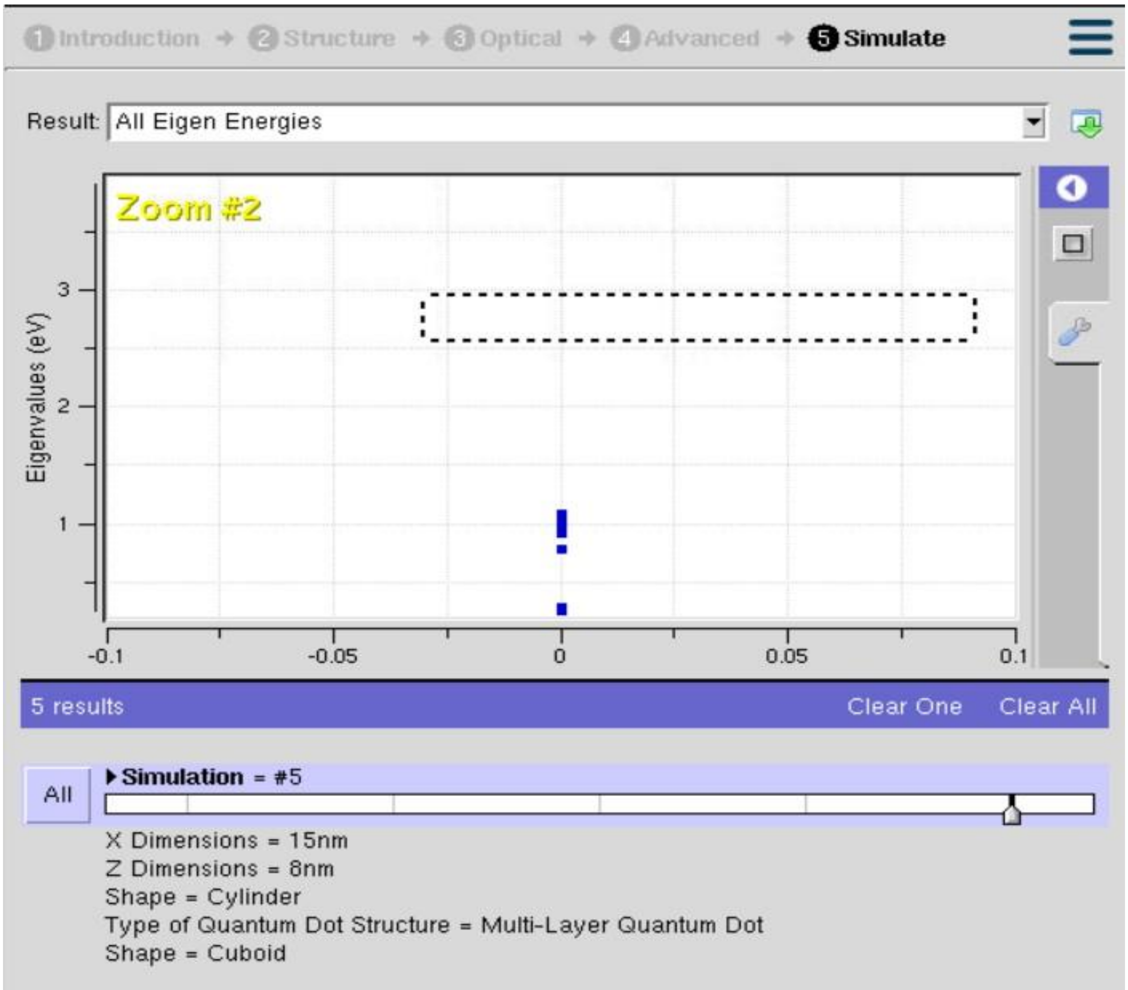
- The graph displays the **allowed energy states (Eigen energies)** of electrons in the quantum dot, measured in **electron volts (eV)**.
- Each **blue square** represents one quantized energy level where an electron can exist.
- The energy levels are **discrete (not continuous)** because electrons are confined in a very small nanoscale structure (quantum confinement).
- The spacing between the energy levels indicates the **energy separation**; larger spacing means stronger quantum confinement.
- These energy levels determine the **electrical and optical properties** of the quantum dot, such as absorption and emission wavelengths.

Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.



- The graph represents the **quantized eigen energy levels** of electrons inside the cylindrical quantum dot.
- Each **blue square** corresponds to one allowed energy state, with values ranging approximately from **1.6 eV to 2.05 eV**.
- The energy levels are **discrete** because electrons are confined within the nanoscale quantum dot.
- The close spacing between some energy levels indicates that electrons can transition between these states with relatively small energy changes.
- These quantized energy levels determine the **electronic and optical behavior** of the quantum dot.

Case Study 2: Simulate the energy states of two layer cuboid quantum dots



- The graph shows the **discrete eigen energy levels** of electrons in the multi-layer quantum dot.
- Each **blue square** represents an allowed energy state, with energy values approximately between **0.3 eV and 1.1 eV**.
- The lower energy range indicates that the multi-layer structure provides **stronger electron confinement** compared to a single-layer quantum dot.
- The spacing between the energy levels reflects the **quantum confinement effect** and the possible electron transition energies.
- These energy states determine the **electronic and optical characteristics** of the quantum dot.

The three graphs show that the eigen energy levels decrease as the quantum dot structure changes from a **basic quantum dot** to a **cylindrical quantum dot** and finally to a **multi-layer cylindrical quantum dot**. The first structure exhibits the highest energy levels (≈2.3–3.8 eV), while the cylindrical structure has intermediate energy levels (≈1.6–2.05 eV). The multi-layer cylindrical quantum dot shows the lowest energy levels (≈0.3–1.1 eV), indicating the strongest carrier confinement. Lower eigen energies and closely spaced levels provide improved control of electron transitions and enhanced optical performance. Therefore, the **multi-layer cylindrical quantum dot** demonstrates the best confinement characteristics, making it the most suitable structure for advanced nanoelectronic and optoelectronic applications.